PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Over-the-counter provision of emergency contraceptive pills: A systematic review
AUTHORS	Atkins, Kaitlyn; Kennedy, Caitlin; Yeh, Ping; Narasimhan, Manjulaa

VERSION 1 – REVIEW

REVIEWER	Kyosuke Kamijo Iida Municipal Hospital
REVIEW RETURNED	22-Jul-2021

GENERAL COMMENTS	Thank you for writing this important original research and for the opportunity to review the manuscript. Evaluating the impact of the availability of emergency contraceptive pills on sexual practices and behavior will help improve sexual and reproductive health and rights of women and girls. There is there is some rationale for the study questions and the manuscript is overall well written. However, the interpretation of the findings is limited by the following issues. Comments and feedback below: General: 1) Throughout the manuscript the terms emergency contraception (EC) and emergency contraceptive pills (ECP) are used confusingly. EC includes ECPs and IUD. I would recommend that you clearly distinguish between these two terms and use them in your manuscript. 2) The risk of bias (RoB) of each the inclusion studies is unclear and therefore, it is difficult to evaluate each outcome. Abstract: There were several reports (these studies are presented below) stating that increasing access to EC did not reduce the number of abortions. Why were the results different in this study? Can you address this in your discussion? Glasier A, Fairhurst K, Wyke S, Ziebland S, Seaman P, Walker J, Lakha F. Advanced provision of emergency contraception does not reduce abortion rates. Contraception. 2004, 69:361-6. Lakha F, Glasier A. Unintended pregnancy and use of emergency contraception among a large cohort of women attending for antenatal care or abortion in Scotland. Lancet. 2006, 368:1782-7. Durrance, Christine, The Effects of Increased Access to Emergency Contraception on Sexually Transmitted Disease and Abortion Rates. Economic Inquiry. 2013, 51: 1682-1695. Hu X, Cheng L, Hua X, Glasier A. Advanced provision of emergency contraception to postnatal women in China makes no difference in abortion rates: a randomized controlled trial. Contraception. 2005, 72:111-6. Strengths and limitations of this study: N/A Introduction: How many countries in the world have OTC ECPs available?

	Line 3-5: Do you have any references? Line5: SRHR is written only two more times in the manuscript. Is there a need to make it an abbreviation? Line33-35: Do you have any references that supply-side constraints and other barriers related to COVID-19 may reduce access and availability of condoms and other forms of medically prescribed contraceptive options? Methods: Line55-56 Sexual and reproductive health has already been described in Line 28 Line52-53: Why are you using Evidence Project eight item risk of bias tool to evaluate non-randomized trials comparative? Cochrane recommends The Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) tool to assess the risk of bias in non-randomized studies of interventions and requires a table to be created to evaluate the study for each outcome. https://training.cochrane.org/handbook/current/chapter-25 P7. Line 12 EC replace with emergency contraceptive Line15-37 Since it was not mentioned in PROSPERO, it would be better to write why you did the additional reviews. Results: N/A Discussion: Abortion There were several reports (these studies are presented above) stating that increasing access to EC did not reduce the number of abortions. Why were the results different in this study? Several limitations on this study should be stated. Conclusion: Is it a conclusion that answers the introduction? Is it appropriate to write an item in conclusion that was not mentioned in PROSPERO? Figure1. Abbreviations should be spelled out. Subtracting exclusion (N=50) from eligibility (N=129) gives N=79, but they do not match (N: 10+56+3=78). Why is that? Table1. The table should be self-explanatory. Abbreviations should be spelled out. Please add a citation to each study. Durrance 2013 †. What † means? Table2. Abbreviations should be spelled out. Using ROBINS-I or Evidence Project eight item risk of bias tool, could you describe the risk of bias for each study for each outcome in this table or in a supplement? Without that, you cannot legitimately assess the reliability of each outcome. Why do you include study (24) in PICO Outcome 3: Unintended Pregnancy? This is because they are looking at contraception rate as an outcome, so it should not exactly be included in Unintended Pregnancy Table3. Please add a citation to each study. Using ROBINS-I or Evidence Project eight item risk of bias tool, could you describe the risk of bias for each study for each outcome in this table? Thank you for the opportunity to review the article and good luck.

REVIEWER	Oskari Heikinheimo University of Helsinki, Obstetrics and gynecology
REVIEW RETURNED	16-Aug-2021

GENERAL COMMENTS	This is a state-of-art review on the effects of OTC-EC. When looking at the individual references, the most recent data is some 10 years

	old, and a lot may have happened during this time. This limitation, of course, is inherit to all research. The findings - limited effects on overall rate of unintended pregnancy / induced abortion, but no increase in sexual risk taking or coercion - have been nicely summarized. However, some of them (especially the decreased rate of induced abortion among adolescents following introduction of OTC-EC, and increased use within 24h of unprotected intercourse) could be highlighted more. One minor specific suggestion - Page 8, line 50/51. 'Finally, evidence from one serial cross sectional study FROM FRANCE showed....'

REVIEWER	Deborah Bateson Family Planning New South Wales
REVIEW RETURNED	29-Oct-2021

GENERAL COMMENTS	Review EC systematic review This is an important systematic review of OTC emergency contraceptive pills which will be useful around the world for policy makers, service deliverers, advocates as well as researchers with an interest in contraception and self-care in relation to supporting a move to OTC provision in their countries. The methodology is appropriate with completion of the PRISMA checklist and the PICO format is well constructed. The systematic review does highlight the challenges of carrying out research in the area of ECPs, as reflected by both the relatively large numbers of studies assessed as having bias and the limitations around the study outcomes. The difference between OTC and behind-the counter access are clearly described. I do have some comments for consideration: 1. Given the data gaps and study challenges in this important area it would be useful for the authors to succinctly summarise future areas for research (and potentially address how researchers could approach this future studies more robustly). Currently the research gaps are scattered throughout the discussion which makes it difficult for readers to grasp them clearly. 2. One of the major gaps, as the authors point out, is lack of research in low resource settings. However, the final sentence in the abstract conclusion is a bit unclear – it states available evidence does not substantively change reproductive health outcomes but evidence gaps remain due to limited studies in low resource settings – does this mean the authors think that this evidence for a positive impact would be provided in these settings and if so this could be explored further in the discussion? This could also be noted in the limitations section (which is currently missing) 3. Subheadings in the discussion would be useful to signpost key areas. I think the case for countries making ECPs OTC (or behind the counter) could be made more clearly as it is challenging for readers to untangle some of the messaging given that mostly it relates to lack of negative impacts by moving from prescription provision rather than positive impacts. 4. The section entitled strengths and limitations of the study is incorrectly headed as it contains broad outcomes of the systematic review rather than strengths and limitations of the current study? The points in this section appear to be key outcomes and they could be made more succinctly and clearly; a section on strengths and limitations of the review is needed?

	5. Dot point 2 in the ‘’strengths and limitations’’ is somewhat confusing as it joins the evidence that there are no substantial changes in EC use, sexual practices and behaviours as a result of OC access (ie a neutral finding) with a positive finding around abortion rates – these should be separated and it could include that this was specifically in a young age subset? 6. In the strengths and limitations (currently incorrectly labelled as such) the statement that EC can prevent up to 95% of pregnancies when taken within 5 days of unprotected intercourse while correct according to WHO is misleading in this context as this figure includes use of a copper IUD which is the most effective method 7. In the introduction lines 22-24 are unclear in relation to studies on OTC oral contraception – as I am not sure how OTC access could result in ‘limited contraindications’ amongst users – does it mean more women used it who had contraindications (the statement re higher continuation is reasonable but was this higher continuation rates?) This sentence needs attention. Thank you for your work on this systematic review and for consideration of these comments for improvement which will help set the case clearly for enhanced access to ECPs.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Thank you for writing this important original research and for the opportunity to review the manuscript. Evaluating the impact of the availability of emergency contraceptive pills on sexual practices and behavior will help improve sexual and reproductive health and rights of women and girls. There is some rationale for the study questions and the manuscript is overall well written. However, the interpretation of the findings is limited by the following issues.

General:

1. Throughout the manuscript the terms emergency contraception (EC) and emergency contraceptive pills (ECP) are used confusingly. EC includes ECPs and IUD. I would recommend that you clearly distinguish between these two terms and use them in your manuscript.

Response: Thank you for noting this. We have reviewed our use of these terms, and decided to exclusively use ECPs, as these are the focus of the review. We have made this change throughout the manuscript.

2. The risk of bias (RoB) of each the inclusion studies is unclear and therefore, it is difficult to evaluate each outcome.

Response: Thank you for this feedback. We discussed risk of bias in-text but had previously omitted it from tables. Per this reviewer’s suggestion below, we have added a column indicating risk of bias for each study described in Table 2. In addition, the full risk of bias tables have been added as an appendix.

Abstract:

1. There were several reports (these studies are presented below) stating that increasing access to EC did not reduce the number of abortions. Why were the results different in this study? Can you address this in your discussion?
    a. Glasier A, Fairhurst K, Wyke S, Ziebland S, Seaman P, Walker J, Lakha F. Advanced provision of emergency contraception does not reduce abortion rates. Contraception. 2004, 69:361-6.
    b. Lakha F, Glasier A. Unintended pregnancy and use of emergency contraception among a large cohort of women attending for antenatal care or abortion in Scotland. Lancet. 2006, 368:1782-7.
    c. Durrance, Christine, The Effects of Increased Access to Emergency Contraception on Sexually Transmitted Disease and Abortion Rates. Economic Inquiry. 2013, 51: 1682-1695.
    d. Hu X, Cheng L, Hua X, Glasier A. Advanced provision of emergency contraception to postnatal women in China makes no difference in abortion rates: a randomized controlled trial. Contraception. 2005, 72:111-6.

Response: Thank you for this important observation. We feel that our review findings are largely in alignment with these studies and note for the reviewer that the above-referenced 2013 paper from Durrance et al. was indeed included in our review. As reported in Table 2, this study found a small but significant decrease in abortion rates following provision of ECPs over the counter. We included four other studies which assessed abortion rates following OTC ECPs, all of which found either no change (aligning with the studies mentioned above) or modest reductions.

Aside from Durrance et al. 2013, the studies mentioned by the reviewer were not included because they did not address over-the-counter ECPs, but their findings provide important context for our review. As such, we have added the following paragraph to our discussion:

> Our finding that OTC ECPs had minimal impacts on unintended pregnancy and abortion may be explained in part by overall low use of ECPs, regardless of conception intention [Glaiser et al 2004; Lakha & Glasier 2006]. However, there is some evidence that even with increased access and uptake, ECPs may have minimal impacts on unintended pregnancy or abortion [Hu 2005]. This should be investigated further; most studies in our review did not report on both pregnancy or abortion and ECP use. In the sole study reporting both unintended pregnancy and ECP uptake [Raine et al 2005], we found no change in either ECP uptake or unintended pregnancy with expanded access. Additional efforts may be required to ensure that increased ECP access reaches those most at risk of unintended pregnancy [Glasier et al 2004].

In the abstract itself, we feel that this is captured with the line “Existing evidence suggests OTC ECPs do not substantively change reproductive health outcomes.” As such, we have not made further changes to the abstract but defer to the editor if additional changes are required.

Introduction:

1. How many countries in the world have OTC ECPs available?

Response: According to the International Consortium for Emergency Contraception, OTC ECPs are available in 19 countries. We have added this to the introduction: “As of December 2021, ECPs are available via pharmacy access in 76 countries, and OTC in 19 countries.”

2. Line 3-5: Do you have any references?

Response: We believe this is in reference to the statement that "reducing barriers to ECPs may increase access to effective contraceptive options, reduce unintended pregnancies, and overall improve sexual and reproductive health and rights (SRHR) outcomes." We have now added a reference to a WHO fact sheet to support this statement.

3. Line 5: SRHR is written only two more times in the manuscript. Is there a need to make it an abbreviation?

Response: We have made this modification; "sexual and reproductive health and rights" is now spelled out in all three instances.

4. Line 33-35: Do you have any references that supply-side constraints and other barriers related to COVID-19 may reduce access and availability of condoms and other forms of medically prescribed contraceptive options?

Response: We have added four references which discuss the impact of COVID-19 on contraceptive choice and availability:

- Aly, J., Haeger, K. O., Christy, A. Y., & Johnson, A. M. (2020). Contraception access during the COVID-19 pandemic. Contraception and reproductive medicine, 5, 17. https://doi.org/10.1186/s40834-020-00114-9
- Haddad LB, RamaRao S, Hazra A, Birungi H, Sailer J. Addressing contraceptive needs exacerbated by COVID-19: A call for increasing choice and access to self-managed methods. Contraception. 2021;103(6):377-9.
- Riley T, Sully E, Ahmed Z, Biddlecom A. Estimates of the Potential Impact of the COVID-19 Pandemic on Sexual and Reproductive Health In Low- and Middle-Income Countries. Int Perspect Sex Reprod Health. 2020;46:73-6.
- Wood, S. N., Karp, C., OlaOlorun, F., Pierre, A. Z., Guiella, G., Gichangi, P., Zimmerman, L. A., Anglewicz, P., Larson, E., & Moreau, C. (2021). Need for and use of contraception by women before and during COVID-19 in four sub-Saharan African geographies: results from population-based national or regional cohort surveys. The Lancet. Global health, 9(6), e793–e801. https://doi.org/10.1016/S2214-109X(21)00105-4

Methods:

1. Line 55-56 Sexual and reproductive health has already been described in Line 28.

Response: Because this is the first use of the acronym SRH, we have kept the text as-is.

2. Line 52-53: Why are you using Evidence Project eight item risk of bias tool to evaluate non-randomized trials comparative? Cochrane recommends The Risk Of Bias In Non-randomized

Studies of Interventions (ROBINS-I) tool to assess the risk of bias in non-randomized studies of interventions and requires a table to be created to evaluate the study for each outcome. https://training.cochrane.org/handbook/current/chapter-25

Response: We are familiar with the ROBINS-I tool and considered it among others (such as the Newcastle-Ottawa scale) when developing the review protocol. The Evidence Project tool differs in several ways from other tools to assess risk of bias in non-randomized or observational studies. First, the Evidence Project tool is applicable to a range of study designs, from case-control studies to cohorts to pe-post studies, which is not the case with some other existing risk of bias tools, many of which are design-specific. Second, it does not need to be adapted for each review, or for each included study. This facilitates comparability across reviews as compared to tools such as ROBINS-I, which asks reviewers to assess bias separately for each outcome included in each study (which may differ across studies and across review topics). As we were doing multiple reviews of different interventions for this WHO self-care guideline, this consistency across reviews was appreciated. Other strengths of the Evidence Project risk of bias tool include its relative ease of use and clarity. The eight items are fairly straightforward and easy to assess, which should make data extraction less prone to error and easier for reviewers with less experience. The tool is also relatively easy for readers to interpret and read, as all information can be condensed into a single table with one row per study. It has also been shown to have moderate to substantial reliability (per the citation included in the manuscript). However, we recognize that the Evidence Project tool is just one among many possible risk of bias tools that systematic reviewers can select from, and does not capture some elements of quality that other tools assess.

3. P7. Line 12 EC replace with emergency contraceptive

Response: This has been updated.

4. Line 15-37 Since it was not mentioned in PROSPERO, it would be better to write why you did the additional reviews.

Response: Thanks for this suggestion. The values and preferences and cost reviews were done to inform required elements of the WHO guideline decision-making process, as specified in the WHO Handbook for Guideline Development. Due to the structure of the PROSPERO database, we focused our reporting there on the primary effectiveness review.

We have now added a sentence to clarify this in the methods section as follows: “These three areas are all required information in the WHO guideline development process.” We have also added a reference to the WHO Handbook for Guideline Development.

Results: N/A

Discussion:

1. There were several reports (these studies are presented above) stating that increasing access to EC did not reduce the number of abortions. Why were the results different in this study?

Response: We have addressed this comment above.

2. Several limitations on this study should be stated.

Response: This has been added at the end of the Discussion under Strengths and Limitations. The new text reads as follows:

> Our review has several strengths and limitations. Our search was comprehensive and included not only literature on the effectiveness of OTC ECPs, but on its costs and the values and preferences of providers and end users. However, we did not include grey literature and conference abstracts, which may have provided valuable information given the evolving nature of this field. We also did not include studies of expanded access to ECPs that did not specifically assess OTC availablity, such as trials of advance provision of ECPs direct to potential users.
>
> Conclusions of this review are also limited by the quality and diversity of included studies. Many observational studies lacked comparison groups or pre/post data, and several did not control for confounding. Further, given the wide range of included study designs and outcomes, we were unable to perform meta-analysis but instead summarized findings qualitatively. Our conclusions are also limited by the concentration of articles in high- and middle-income settings; future research should examine the impacts of OTC ECPs in resource-limited settings.

Conclusion:

1. Is it appropriate to write an item in conclusion that was not mentioned in PROSPERO?

Response: As mentioned above, while not included in PROSPERO due to the lack of relevant fields for complementary reviews, the values and preferences and costs reviews were developed a priori and included in our overall protocol, and they informed required elements of the WHO guideline decision-making process. The conclusion that “providing ECPs OTC may be cost-saving and responsive to user preferences” was raised as important when we presented our findings to stakeholders. We have included it here so that readers looking for this information can easily find it, but defer to the editor if this should be removed from the conclusion.

Figure 1:

1. Abbreviations should be spelled out.

Response: This has been revised throughout the manuscript.

2. Subtracting exclusion (N=50) from eligibility (N=129) gives N=79, but they do not match (N: 10+56+3=78). Why is that?

Response: The 79 included articles came from 77 total studies. Of these 79 articles, three were solely included in the cost review, 20 solely in the PICO review, 55 solely in the values and preferences review, and one in both the PICO and values & preferences reviews: 3+20+55+1=79. We hope this clarifies; we tried to explain this in the figure using language such as "N=19 studies in 21 articles" and "N=56, including one in PICO."

Tables:

1. Abbreviations should be spelled out.

Response: We have reviewed the table and spelled out most uncommon abbreviations (e.g. NHANES, NR, NS, OC). Given the frequency with which "OTC" and "ECP" appear in the text and tables, we have retained these abbreviations to improve table readability.

2. Please add a citation to each study.

Response: This has been updated.

3. Table 1: Durrance 2013 †. What † means?

Response: Thank you for flagging this; the footnote symbol was an error and we have deleted it.

1. Tables 2-3: Using ROBINS-I or Evidence Project eight item risk of bias tool, could you describe the risk of bias for each study for each outcome in this table or in a supplement? Without that, you cannot legitimately assess the reliability of each outcome.

Response: We have added this for Table 2. We note that we only conducted risk of bias assessments for studies included in the main effectiveness (PICO) review.

2. Table 2: Why do you include study (24) in PICO Outcome 3: Unintended Pregnancy? This is because they are looking at contraception rate as an outcome, so it should not exactly be included in Unintended Pregnancy.

Response: Thank you for highlighting this important point. We have rephrased the discussion of this study in the narrative above the table as follows:

"Additionally, one ecological study assessed changes in conception rate over time in the UK [24], which does not explicitly consider whether the pregnancy was intended but is an indirect proxy measure. The study found…"

In addition, we have inserted the below footnote in Tables 1 and 2:

"This study assessed changes in conception rate, which does not explicitly consider whether the pregnancy was intended but is considered an indirect proxy measure."

Thank you for the opportunity to review the article and good luck.

Reviewer 2

This is a state-of-art review on the effects of OTC-EC. When looking at the individual references, the most recent data is some 10 years old, and a lot may have happened during this time. This limitation, of course, is inherit to all research.

1. The findings - limited effects on overall rate of unintended pregnancy / induced abortion, but no increase in sexual risk taking or coercion - have been nicely summarized. However, some of them (especially the decreased rate of induced abortion among adolescents following introduction of OTC-EC, and increased use within 24h of unprotected intercourse) could be highlighted more.

Response: Thank you for this positive feedback. We have made some revisions to our discussion which we feel better emphasize these positive findings:

"Though we found minimal changes in overall ECP use in OTC models, two studies included in the review found that after OTC provision, use of ECPs within 24 hours of sex increased [29, 38]. This is promising, given ECPs are more effective when used promptly.

For most outcomes, our review did not identify any substantial or concerning differences by age. However, there is promising evidence regarding OTC ECPs among younger women. Observational evidence included in our review showed that abortion rates decreased significantly among younger age groups…"

We have further emphasized this through revisions to the abstract:

Results: "… Broadly, there were no differences in overall ECP use, pregnancy, or sexual behavior, and an increase in timely ECP use, when comparing …"

Conclusion: "OTC ECPs are feasible and acceptable. They may increase access to and timely use of effective contraception …"

2. One minor specific suggestion - Page 8, line 50/51. 'Finally, evidence from one serial cross sectional study FROM FRANCE showed....'

Response: Thank you for this suggestion; we have revised the statement:

"Finally, evidence from one serial cross-sectional study from France showed that reporting ever having an abortion declined from 17.0% before OTC ECP access to 15.6% after OTC ECP access (p=0.04) [28]."

Reviewer 3

This is an important systematic review of OTC emergency contraceptive pills which will be useful around the world for policy makers, service deliverers, advocates as well as researchers with an interest in contraception and self-care in relation to supporting a move to OTC provision in their countries.The methodology is appropriate with completion of the PRISMA checklist and the PICO format is well constructed. The systematic review does highlight the challenges of carrying out research in the area of ECPs, as reflected by both the relatively large numbers of studies assessed as having bias and the limitations around the study outcomes. The difference between OTC and behind-the counter access are clearly described. I do have some comments for consideration.

1. Given the data gaps and study challenges in this important area it would be useful for the authors to succinctly summarise future areas for research (and potentially address how researchers could approach this future studies more robustly). Currently the research gaps are scattered throughout the discussion which makes it difficult for readers to grasp them clearly.

Response: Thank you for this suggestion. We have rephrased the discussion to include a dedicated section on areas for future research, which includes the suggestions previously scattered throughout the discussion. We hope this improves readability. The new section reads as follows:

Our review highlights some critical areas for future research on OTC ECPs. First, given provider concerns identified in our review, future research should also assess whether correct knowledge of ECPs translates to correct use in OTC modalities.

Second, future research should more closely examine the impact of OTC ECPs among key subgroups, including younger women. This is important as self-care interventions may present unique opportunities and challenges for different populations and in different settings [7]. For example, it is unclear through what mechanisms OTC ECPs may differentially impact outcomes such as abortion or STI acquisition among younger age groups, and if routine preventive SRH care plays a role. Equitable implementation of OTC ECPs as a self-care intervention should consider the intersecting roles of race/ethnicity/culture/language, occupation, gender/sex, religion, education, health literacy, socioeconomic status, and social capital as determinants of sexual and reproductive health and rights and key factors affecting delivery, uptake, and impact of OTC ECPs [108].

Finally, though OTC ECPs are an important contraceptive option for individuals, communities, and health systems worldwide, the evidence base identified through our effectiveness, cost, and values and preferences reviews was concentrated in high-income settings. Specifically, we only found evidence of OTC ECPs' effectiveness and costs from high-income countries. In our values and preferences review, 80% of identified studies were from high-income settings, and a low-income setting (Democratic Republic of Congo) was represented in a single study [66]. Meaningful efforts are needed to recognize, invest in, and promote future research on the effects of increased OTC ECPs in low- and middle-income countries. Future research

should particularly consider impacts on user cost in these settings, given concerns identified in this review.

2. One of the major gaps, as the authors point out, is lack of research in low resource settings. However, the final sentence in the abstract conclusion is a bit unclear – it states available evidence does not substantively change reproductive health outcomes but evidence gaps remain due to limited studies in low resource settings – does this mean the authors think that this evidence for a positive impact would be provided in these settings and if so this could be explored further in the discussion? This could also be noted in the limitations section (which is currently missing)

Response: Thanks for this thoughtful question. Our intent with this sentence was to summarize the evidence base while acknowledging its limitations. We have divided this into two sentences to clarify, and have noted other limitations that are summarized in our revised discussion:

> "Existing evidence suggests OTC ECPs do not substantively change reproductive health outcomes. Future studies should examine OTC ECPs' impacts on user costs, among key subgroups, and in low-resource settings."

3. Subheadings in the discussion would be useful to signpost key areas. I think the case for countries making ECPs OTC (or behind the counter) could be made more clearly as it is challenging for readers to untangle some of the messaging given that mostly it relates to lack of negative impacts by moving from prescription provision rather than positive impacts.

Response: Thanks for this. We have included the following subheadings to organize the discussion more clearly: (1) OTC ECP effectiveness, (2) areas for future research, and (3) strengths and limitations.

4. The section entitled strengths and limitations of the study is incorrectly headed as it contains broad outcomes of the systematic review rather than strengths and limitations of the current study? The points in this section appear to be key outcomes and they could be made more succinctly and clearly; a section on strengths and limitations of the review is needed?

Response: Thank you for this comment. We have reviewed BMJ Open guidelines for this section, and agree with the author that the section as written overly focused on results, rather than on strengths and limitations of the study methods for the systematic review. We have revised it accordingly:

- We comprehensively searched the literature on effectiveness, costs, and values and preferences of over-the-counter emergency contraception.
- We searched four major databases and four clinical trial registries, with no restrictions on language or geography.
- Given our focus on over-the-counter delivery modalities, we may have excluded studies that assessed relevant outcomes of expanded access to emergency contraception through advance provision or other modalities.
- The findings of this review may not be generalizable, as the majority of studies were conducted in high- and middle-income countries.

We have also added a section to the Discussion summarizing strengths and limitations of the review in more detail.

5. Dot point 2 in the ‘’strengths and limitations’’ is somewhat confusing as it joins the evidence that there are no substantial changes in EC use, sexual practices and behaviours as a result of OC access (ie a neutral finding) with a positive finding around abortion rates – these should be separated and it could include that this was specifically in a young age subset?

Response: This comment refers to text which has been deleted given the decision to revise this section in according with the reviewer’s previous comment and BMJ Open guidelines.

6. In the strengths and limitations (currently incorrectly labelled as such) the statement that EC can prevent up to 95% of pregnancies when taken within 5 days of unprotected intercourse while correct according to WHO is misleading in this context as this figure includes use of a copper IUD which is the most effective method.

Response: This comment refers to text which has been deleted given the decision to revise this section in according with the reviewer’s previous comment and BMJ Open guidelines.

7. In the introduction lines 22-24 are unclear in relation to studies on OTC oral contraception – as I am not sure how OTC access could result in ‘limited contraindications’ amongst users – does it mean more women used it who had contraindications (the statement re higher continuation is reasonable but was this higher continuation rates?) This sentence needs attention.

Response: Thanks for this; the term “contraindications” was used erroneously and we appreciate the reviewer bringing it to our attention. In the cited review, one study found that while OTC users were more likely to report a WHO category 3 contraindication, there was no difference in category 4 contraindications; another study found no differences in contraindicated contraceptive use between OTC and non-OTC users. We have revised the sentence to clarify this.

The revised sentence now reads: “…OTC oral contraception may result in higher continuation with limited contraindicated use among users.”

Thank you for your work on this systematic review and for consideration of these comments for improvement which will help set the case clearly for enhanced access to ECPs.